

FAHUD, Oman

WMO: 412620

Lat: 22.3475N

Lon: 56.4903E

Elev: 168

StdP: 99.32

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	12.5	-7.4	2.0	29.4	-5.2	2.5	30.3	11.9	22.6	11.0	22.4	2.6	270	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	46.6	21.3	45.6	21.4	44.5	21.3	25.2	35.8	24.8	36.1	24.4	35.9	3.6	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.0	18.1	29.2	22.2	17.2	29.2	21.4	16.4	29.4	78.6	36.5	76.8	35.7	75.0	35.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	10.0	8.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.4	19.9	21.8	25.5	30.2	35.0	36.8	37.2	35.6	33.4	30.1	25.2	21.6
	DBStd	6.50	2.11	2.75	3.02	2.77	2.08	1.84	2.01	2.41	1.91	2.03	2.05	2.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	14	8	3	0	0	0	0	0	0	0	0	0	3
	CDD10.0	7086	307	332	480	606	776	804	843	795	702	624	456	361
	CDD18.3	4058	57	101	222	356	518	554	584	537	452	366	206	106
	CDH23.3	63295	473	1052	2663	5228	8783	9735	10400	9212	7303	5348	2221	876
CDH26.7	42980	70	356	1337	3294	6368	7340	7927	6775	4994	3320	985	213	
Wind	WSAvg	3.5	3.0	4.0	4.2	3.4	4.0	4.1	3.9	4.2	3.9	2.5	2.3	3.0
Precipitation	PrecAvg	75	9	10	13	9	3	5	5	5	3	2	4	7
	PrecMax	138	19	34	93	57	20	36	22	18	21	6	23	30
	PrecMin	28	3	1	1	1	0	0	1	1	0	0	0	1
	PrecStd	29	4	7	18	12	5	7	6	4	4	2	5	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	34.6	37.6	42.0	47.0	48.0	47.5	46.7	44.0	41.1	35.2	31.4
		MCWB	16.4	17.5	17.6	19.1	20.6	20.9	22.1	21.8	20.8	19.2	18.0	17.0
	2%	DB	28.4	32.2	35.7	40.4	44.9	46.7	46.4	45.4	42.6	39.7	34.0	29.9
		MCWB	16.4	16.8	17.6	18.9	20.1	20.8	21.9	22.1	20.9	19.2	17.9	17.2
	5%	DB	27.2	30.3	34.4	39.1	43.4	45.4	45.4	44.2	41.5	38.4	32.9	29.0
		MCWB	16.2	16.5	17.5	18.7	19.9	20.7	21.9	22.3	21.0	19.1	18.1	17.4
	10%	DB	26.1	28.8	33.1	37.8	42.2	44.2	44.3	43.0	40.2	37.1	31.6	27.8
		MCWB	15.6	16.3	17.4	18.2	19.8	20.6	21.9	22.3	20.9	18.8	17.9	17.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	20.1	23.0	24.3	25.2	26.0	25.8	25.5	24.6	24.2	22.4	20.5
		MCDB	23.5	26.1	27.7	30.5	32.3	33.1	37.8	38.6	34.7	31.7	28.0	24.5
	2%	WB	18.9	19.3	21.8	22.9	24.5	25.2	25.2	24.7	24.0	23.3	21.2	19.7
		MCDB	24.2	26.5	27.1	29.9	32.2	34.0	37.6	38.3	34.8	31.5	27.5	25.2
	5%	WB	17.9	18.6	20.9	21.9	23.6	24.7	24.7	24.3	23.6	22.4	20.4	19.0
		MCDB	24.6	26.5	27.5	30.2	33.1	34.5	37.3	37.5	34.1	31.4	27.7	25.5
	10%	WB	16.9	17.8	20.0	20.9	22.6	24.1	24.4	23.9	23.1	21.3	19.6	18.1
		MCDB	24.3	26.0	27.5	30.6	35.0	35.0	36.9	36.8	33.3	31.5	27.9	25.9
Mean Daily Temperature Range	5% DB	MDBR	13.1	14.2	14.0	14.4	14.8	15.1	14.4	14.1	14.0	14.6	13.4	13.0
		MCDBR	13.8	15.3	15.2	15.3	15.9	16.3	15.2	14.7	15.1	15.4	14.2	13.6
		MCWBR	6.0	6.2	5.7	5.0	5.2	4.9	4.2	3.2	3.9	5.3	5.9	5.8
	5% WB	MCDBR	12.0	13.1	11.4	12.3	13.0	14.1	14.0	13.8	13.5	12.8	12.1	12.5
		MCWBR	5.9	5.9	5.0	5.3	5.0	4.6	3.4	2.8	3.1	5.2	5.8	6.2
Clear-Sky Solar Irradiance	taub		0.417	0.466	0.527	0.584	0.622	0.709	0.795	0.760	0.636	0.490	0.435	0.412
	taud		2.082	1.926	1.774	1.685	1.616	1.495	1.408	1.455	1.623	1.896	2.050	2.091
	Ebn at Noon		836	817	786	749	717	651	598	619	690	784	810	823
	Edh at Noon		148	184	224	249	265	297	324	309	257	187	152	142
All-Sky Solar Radiation	RadAvg		4.79	5.57	6.37	6.98	7.51	7.35	6.69	6.72	6.44	5.93	5.01	4.54
	RadStd		0.16	0.18	0.30	0.34	0.15	0.26	0.27	0.16	0.10	0.15	0.08	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page